Laboratory report of Nanjing University of Information Science and Technology

Experiment Title Learn AWS Data Class Applied Computing Name Lu Moyou ID 202483910013

一、Experimental purpose

（1）Learn basic knowledge about AWS

（2）Learn to use instances in AWS

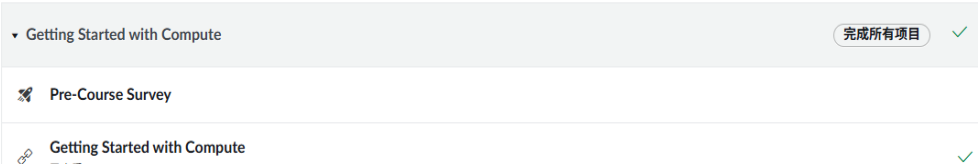
二、Experiment Contents

- Launch an EC2 instance with termination protection turned on.
- Monitor your EC2 instance.
- Modify the security group that your web server is using to allow HTTP access.
- Connect to your EC2 instance using AWS Systems Manager Fleet Manager.
- Manage the state of an EC2 instance.
- Change your EC2 instance type.
- Test termination protection.
- Explore Amazon EC2 limits.

三、Detailed steps of the experiment（please show the experiment steps and screenshots of the results）

Tasks:

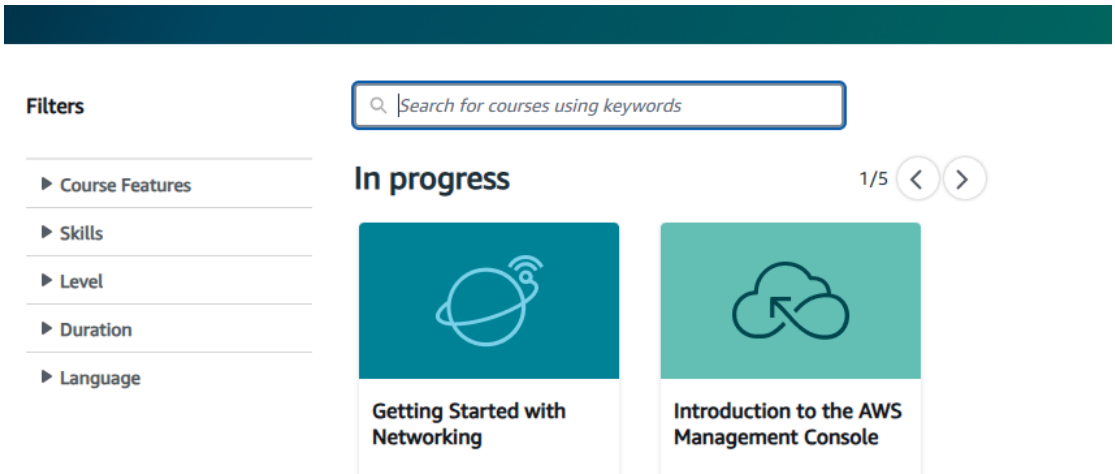
1. Watch: Getting Started with Compute



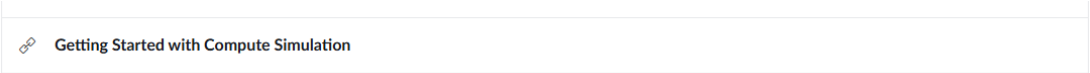
First, register with AWS Educate, then just click the link above.

If you cannot find the course from the above link, try the link below:

Learner Dashboard (https://www.awseducate.com/student/s/content) and search the course: Getting Started with Compute



2. Getting Started with Compute Simulation

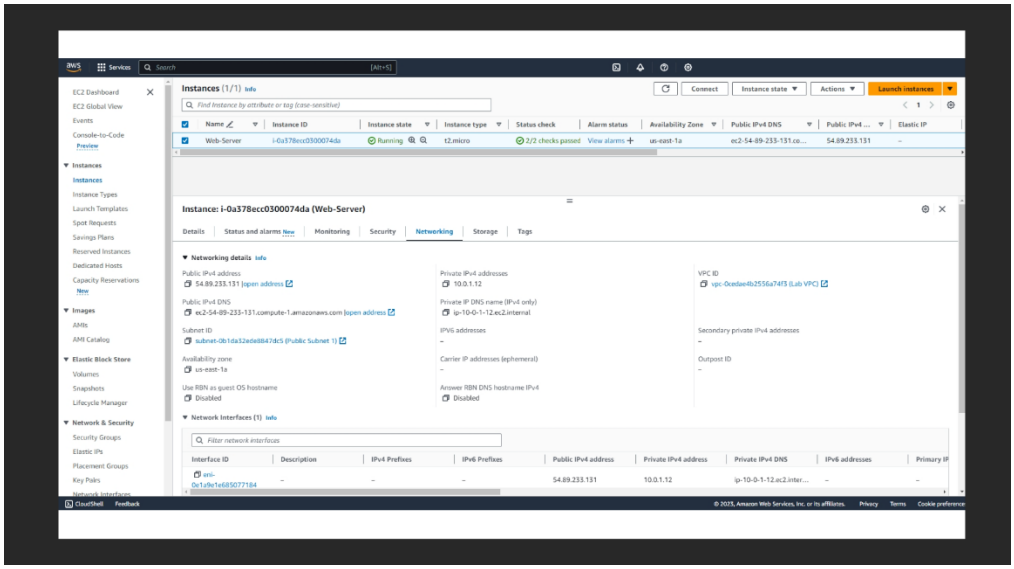


Please click the link above to complete your tasks. The experimental steps can be found in the instruction file attached.

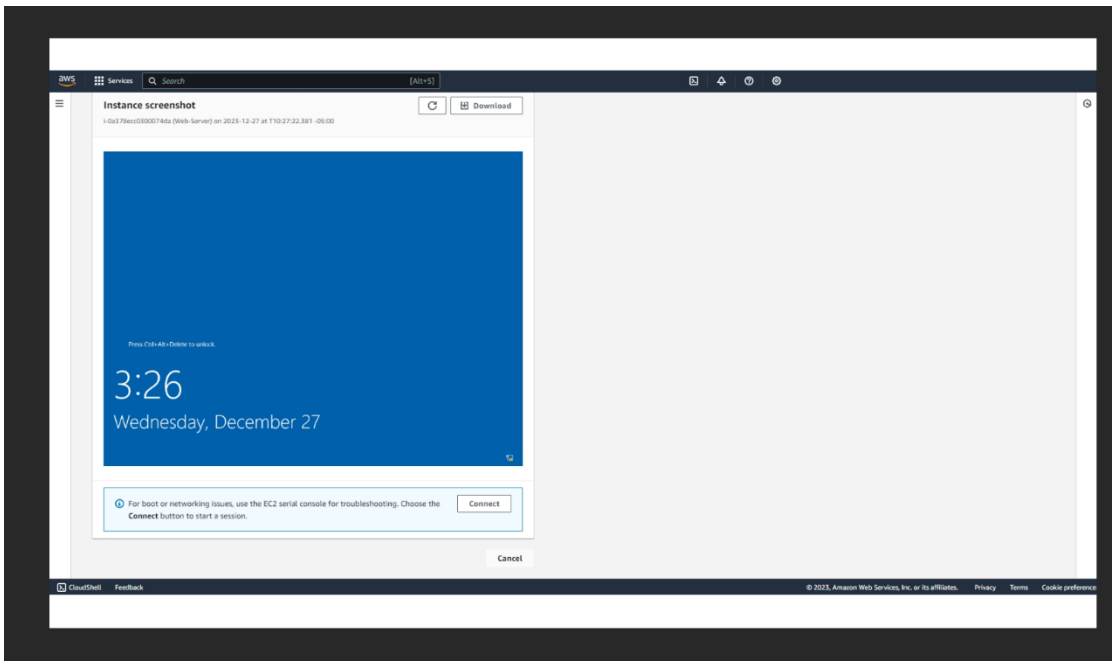
四、Reflection

Screen shots:

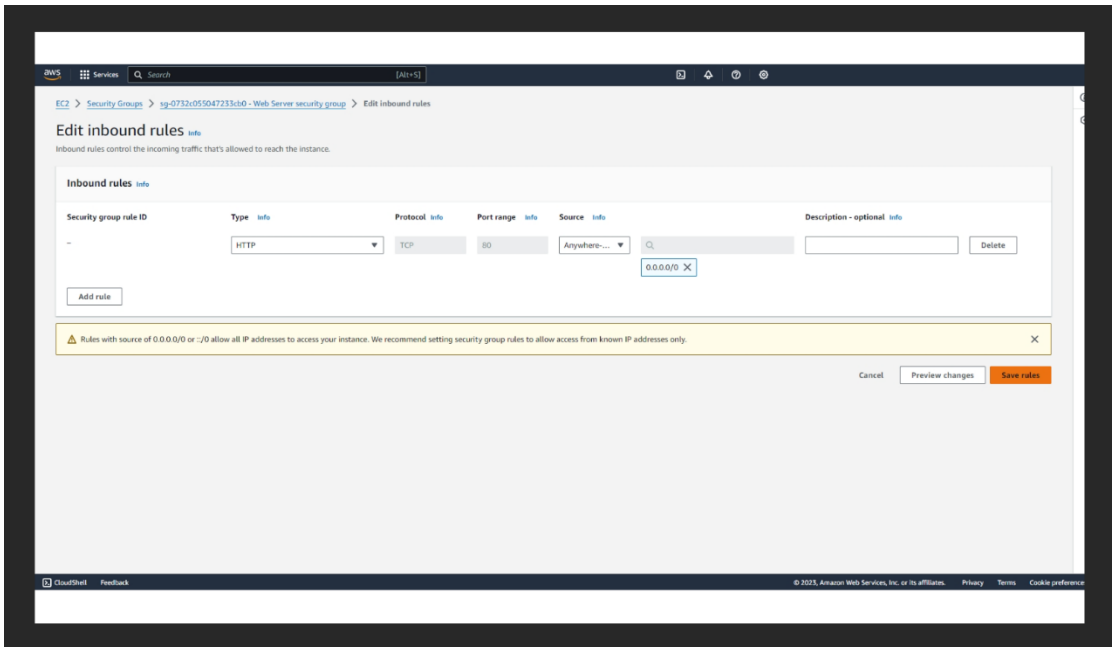
Task1:


Instances (1/1)
Instance: i-0a378ecc0300074da (Web-Server)
Networking
Network Interfaces (1)

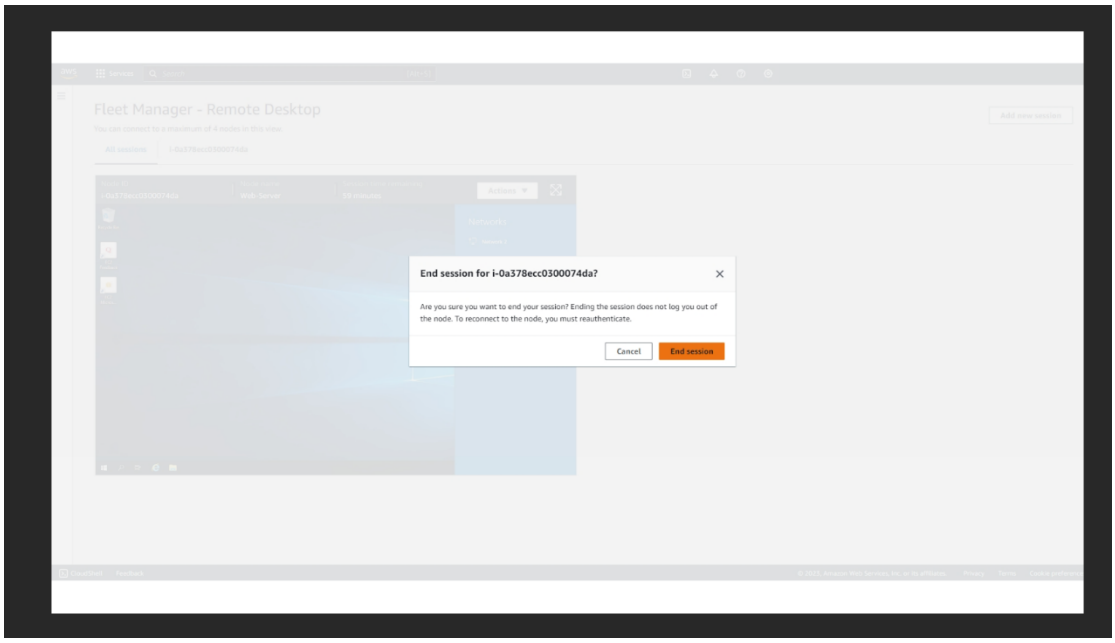
Task2：


Instance screenshot
Download
3:26
Wednesday, December 27
Connect
Cancel

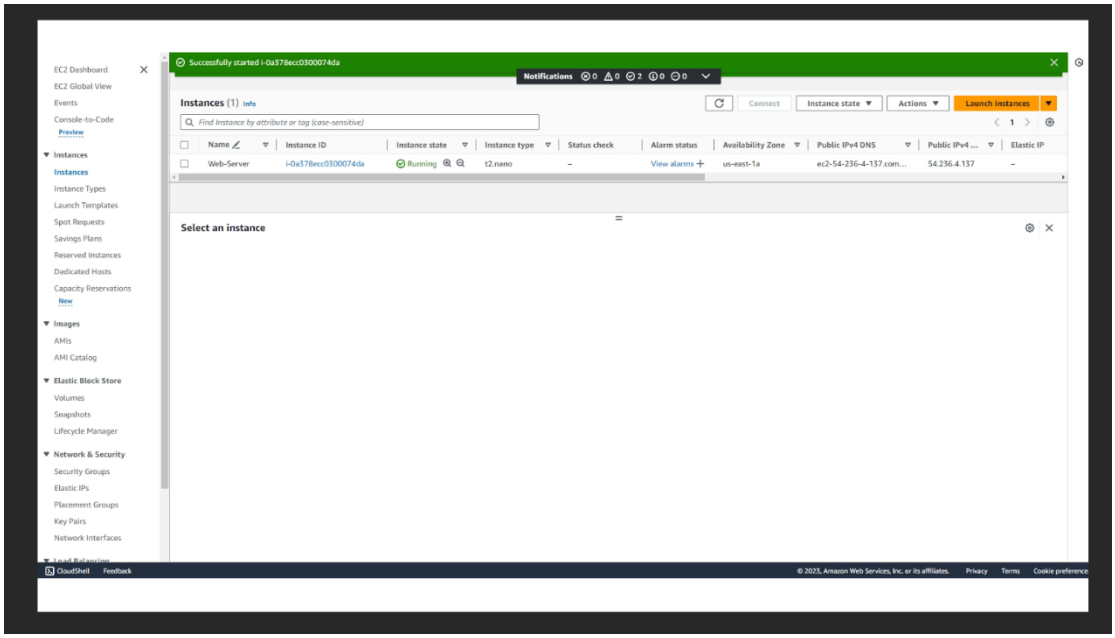
Task3：


Services
Search
[Alt+S]
EC2 > Security Groups > sg-0732c055047233cb0 - Web Server security group > Edit inbound rules
Edit inbound rules Info
Inbound rules control the incoming traffic that's allowed to reach the instance.
Inbound rules Info
Security group rule ID
Type Info
Protocol Info
Port range Info
Source Info
Description - optional Info
-
HTTP
TCP
80
Anywhere-...
0.0.0.0/0
Delete
Add rule
Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.
Cancel
Preview changes
Save rules
CloudShell
Feedback
© 2023, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preference

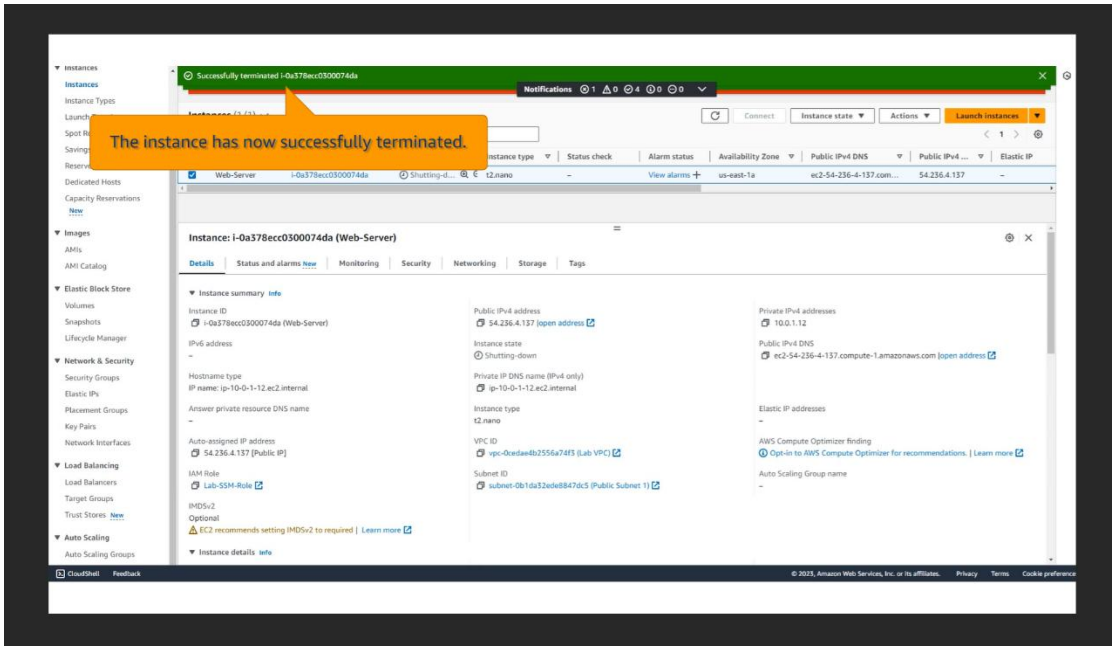
Task4：


Fleet Manager - Remote Desktop
End session for i-0a378ecc0300074da?
Are you sure you want to end your session? Ending the session does not log you out of the node. To reconnect to the node, you must reauthenticate.
Cancel
End session

Task5：



Task6：



1. What is an EC2 instance essentially?

An EC2 instance is essentially a virtual server in the cloud provided by the Amazon EC2 web service, which offers resizable compute capacity.

2. What is the main advantage of server virtualization?

The main advantage is that it provides flexible, resizable compute capacity and lets you quickly scale resources up or down to match changing computing requirements, without needing physical hardware.

3. **Why can cloud computing create servers quickly?**

   Cloud computing (Amazon EC2) can create servers quickly because it uses pre-configured Amazon Machine Images (AMIs) and automated provisioning; you don't need to buy, install, or configure physical hardware, and instances can be launched and booted in minutes via the cloud platform.